

Thakur Degree College of Science and Commerce

KANDIVALI (EAST)

MUMBAI

A

PROJECT REPORT

ON

Skill-Edge

For

THAKUR COLLEGE OF SCIENCE & COMMERCE

Shyamnarayan Thakur Marg, Thakur Village, Kandivali (E), Mumbai - 400 101

Designed and Developed

BY: **Mr. Vivek Vinod Gupta**

Submitted in partial fulfillment of

Bachelors of Science (Computer Science)

[UNIVERSITY OF MUMBAI]

Thakur Degree College of Science and Commerce

Kandivali (East), Mumbai.

ACADEMIC YEAR 2024-2025

CERTIFICATION FROM COLLEGE



Thakur Degree College of Science & Commerce
Shyamnarayan Thakur Marg, Thakur Village
Kandivali (E), Mumbai - 400 101

DATE: _____

COMPUTER SCIENCE DEPARTMENT

(2024-2025)

Certificate of Approval

This is to certify that the project work entitled **“Skill-Edge”** is prepared by **Mr. Vivek Vinod Gupta** a student of **“Third Year Bachelor Of Science (Computer Science)”** course of University of Mumbai, which is conducted by our college.

This is the original study work and important sources used have been duly acknowledged in the report. The report is submitted in partial fulfillment of B.Sc. (Computer Science) course as per rules of University of Mumbai.

Project Guide

Head of Department

External Examiner

ACKNOWLEDGEMENT

ACKNOWLEDGEMENT

Achievement is finding out what you would be doing rather than what you have to do. It is not until you undertake such a project that you realize how much effort and hard work it really is, what are your capabilities and how well you can present yourself or other things. It tells us how much we rely on the efforts and goodwill of others. It gives me immense pleasure to present this report towards the fulfillment of my project.

It has been rightly said that we are built on the shoulder of others. For everything I have achieved, the credit goes to all those who had helped me to complete this project successfully.

I take this opportunity to express my profound gratitude to management of Thakur Degree College Of Science & Commerce for giving me this opportunity to accomplish this project work.

I am very much thankful to **Dr. C.T.Chakraborty**- Principal of Thakur College for their kind co-operation in the completion of my project.

A special vote of thanks to my faculty, **Mr. Ashish Trivedi** who is our HOD & also our project guide **Mrs. Riddhi Mishra** for their most sincere, useful and encouraging contribution throughout the project span, without them we couldn't start and complete the project on time.

Finally, I would like to thank all my friends & entire Computer Science department who directly or indirectly helped me in completion of this project & to my family without whose support, motivation & encouragement this would not have been possible.

(VIVEK VINOD GUPTA)

INDEX

INDEX

Sr. No.	Index Topic	Page
	Acknowledgement	7
I.	Introduction	
(i)	Description of System	10
(ii)	Limitations of present system	10
(iii)	Proposed system and its advantages	11
(iv)	Requirements Specification	12
(v)	Activity Chart	14
II	System Analysis	16
(i)	Event Table	18
III	UML Diagram	21
(i)	Use Case Diagram	22
(ii)	Component Diagram	23
(iii)	Deployment Diagram	23
(iv)	State Diagram	24
(v)	Class Diagram	25
(vi)	Collabration Diagram	25
(vii)	Activity Diagram	26
(viii)	Sequence Diagram	28

IV	Data Flow Diagram	
(i)	Data Flow Diagram(DFD)	30
V	System Implementation	31
(i)	Code Implementation	32
VI	Results	37
(i)	Test Cases	38
(ii)	Database Structure	41
(iii)	ScreenShorts	46
VII	Future Scope	51
VIII	Conclusion	53
IX	References and Bibliography	55

INTRODUCTION

Description of System

SkillEdge is a comprehensive AI-driven web application designed to analyze student performance and provide personalized guidance. Built with Django, HTML, CSS, and JavaScript, the system offers features like secure JWT-based authentication, performance analysis, AI-driven recommendations, and progress tracking through interactive data visualizations. Students can input academic data, receive customized study plans, and monitor their growth on dynamic dashboards. The responsive design ensures a seamless experience across devices, and the platform also fosters community engagement by connecting students with peers and mentors for collaborative learning.

Limitations of Present System

✗ **Manual Data Entry:** Requires manual input of performance data, which can lead to errors and reduced efficiency.

✗ **Limited Predictive Analytics:** The system may not utilize advanced machine learning models for deeper predictive insights.

✗ **Lack of Integration:** Limited integration with external learning resources or APIs to enhance guidance.

✗ **No Mobile App:** The absence of a dedicated mobile application limits accessibility on smartphones and tablets.

✗ **Scalability Concerns:** Potential performance and scalability issues if the user base grows significantly.

✗ **User Engagement:** The community guidance feature might not be fully developed, impacting collaborative learning opportunities.

Proposed System

The proposed **SkillEdge** system aims to enhance the current platform by introducing advanced features and addressing existing limitations. The new system will automate data input through tools like OCR and spreadsheet import, reducing manual errors.

It will integrate advanced machine learning models to provide predictive analytics and personalized learning insights. External APIs from educational platforms (e.g., Coursera, Khan Academy) will be integrated to offer tailored study resources. A dedicated mobile app for both Android and iOS will improve accessibility and user engagement. The system will leverage cloud services (e.g., AWS, Azure) to ensure scalability and performance for a larger user base. Enhanced community features, including forums and mentorship programs, will promote collaborative learning.

Real-time data visualization with libraries like Plotly or D3.js will provide dynamic insights, while gamification elements such as rewards and progress milestones will keep students motivated throughout their learning journey.

Advantages of Proposed System

- ✓ **Improved Data Accuracy:** Automated data input reduces manual errors, enhancing the reliability of performance analysis.
- ✓ **Enhanced Predictive Analytics:** Advanced machine learning models provide accurate performance predictions and personalized guidance.
- ✓ **Resource Integration:** External APIs offer tailored learning resources, enriching the student experience.
- ✓ **Increased Accessibility:** The mobile app ensures users can access the platform on the go, improving engagement.
- ✓ **Scalability and Performance:** Cloud integration supports a growing user base with stable and fast performance.

REQUIREMENT SPECIFICATION

1. Hardware Requirements

- **Processor:** Modern multi-core processor (e.g., Intel Core i5 or equivalent)
- **RAM:** Minimum of 8 GB
- **Storage:** SSD with at least 50 GB of free space for development and testing
- **Network:** Stable internet connection for development, dependency downloads, and version control

2. Software Requirements

- **Operating System:** Windows, macOS, or Linux
- **Python Version:** 3.9 or higher
- **Django Version:** 4.0 or higher
- **Database:** PostgreSQL, MySQL, or SQLite (depending on configuration)
- **Web Browser:** Modern browser for testing (e.g., Google Chrome, Mozilla Firefox)
- **IDE/Text Editor:** Visual Studio Code, PyCharm, or any preferred code editor

3. Front-End Technologies

- **HTML/CSS:** For structuring and styling the web pages
- **JavaScript:** For interactive elements and dynamic content
- **Bootstrap:** For responsive design and UI components
- **Charts Library:** For data visualization and performance analytics

4. Back-End Technologies

- **Django:** Framework for developing the SkillEdge web application
- **Database:** SQLite for lightweight and efficient data management
- **JWT Authentication:** For secure user login and session management
- **Django Apps:** Modular apps for authentication, performance analysis, prediction, and community interaction

ACTIVITY CHART

THAKUR COLLEGE OF SCIENCE AND COMMERCE

Department of Computer Science

2024-2025

Student's Name: Gupta Vivek Vinod

Project Name: Skill-Edge

College Name: Thakur College of Science and Commerce.

PHASES	EXPECTED DATE OF COMPLETION	ACTUALTIMIN G OF COMPLETION	SIGNATURE
Preliminary Investigation	23/11/24	23/11/24	
System Analysis	07/12/24	14/12/24	
System Designing	21/12/24	04/01/25	
System Coding	25/01/25	08/01/25	
System Implementation	15/01/25	17/01/25	
Testing Phase	20/03/25	20/03/25	
Project Submission			

SYSTEM ANALYSIS

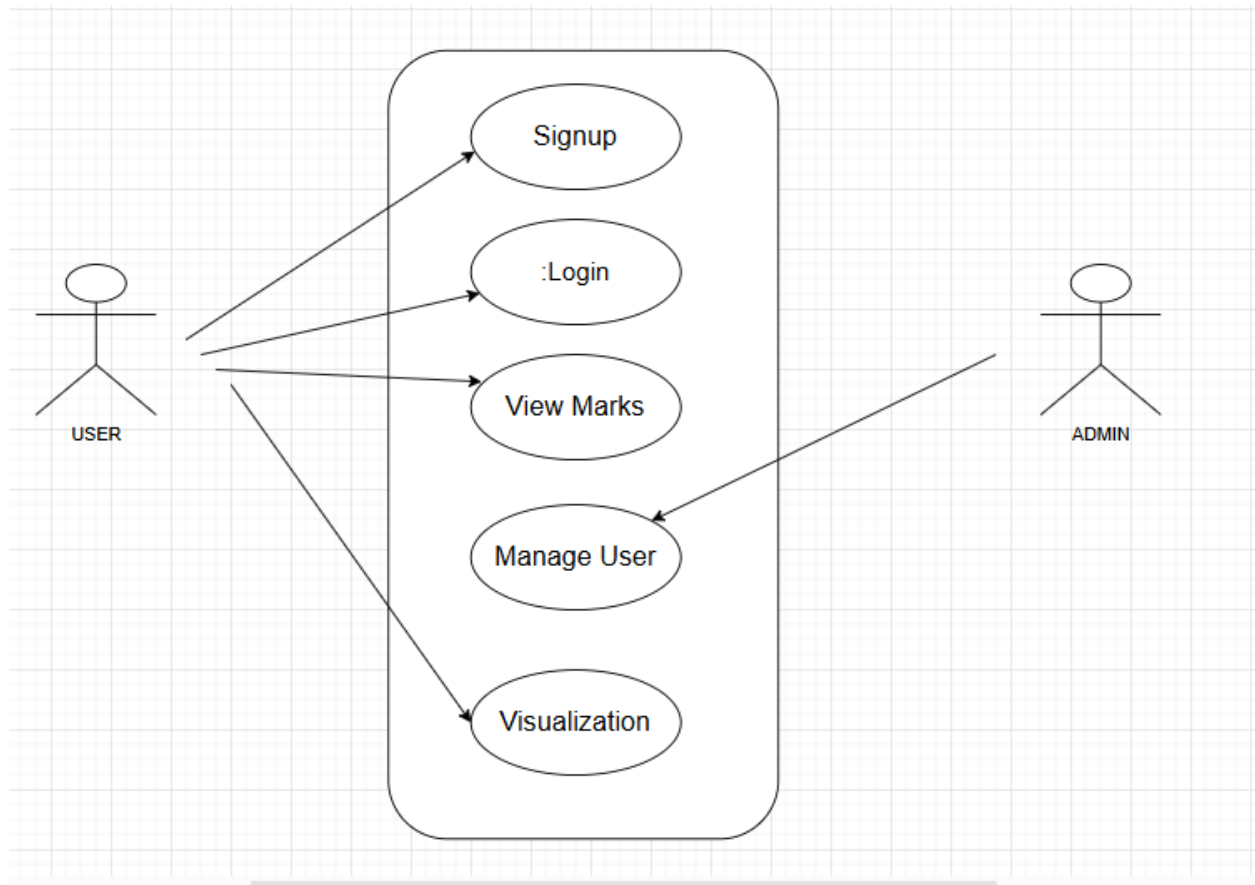
EVENT TABLE

EVENT TABLE

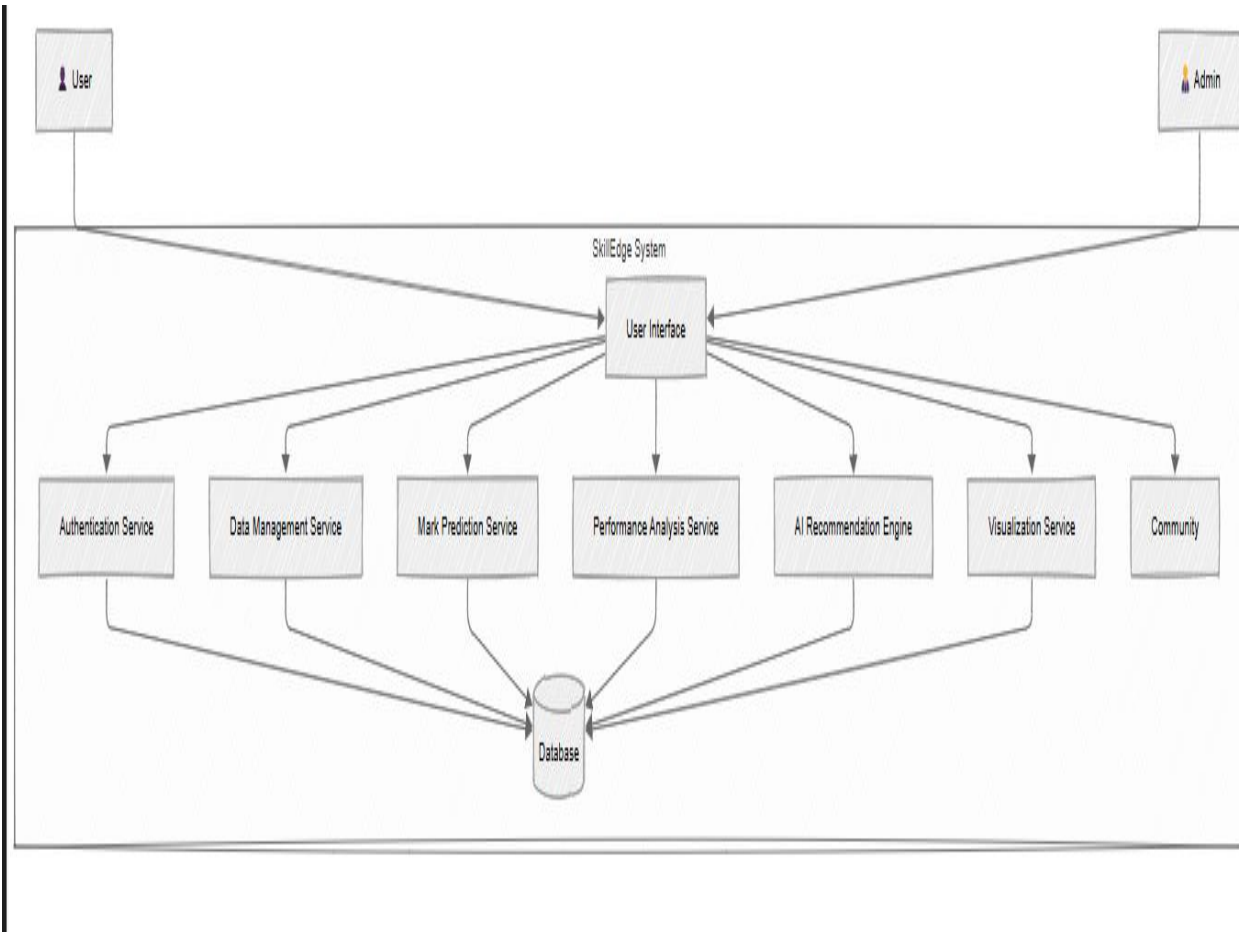
ID	Description	Trigger	Response
1	Sign Up	Submit Signup Form	Go to Login
2	Login	Submit Login Form	Go to Dashboard
3	Data Entry	Input Academic Data	Data Reflect
4	Marks Prediction	Request Mark Prediction	Display Predicted Marks
5	Performance Analysis	Request Analysis	Show Charts and analysis
6	AI Recommendation	Analysis Completed	Study Tips
7	Visualization	View Analytics	Visual Insights
8	Logout	Logout	Redirect to login

UML DIAGRAM

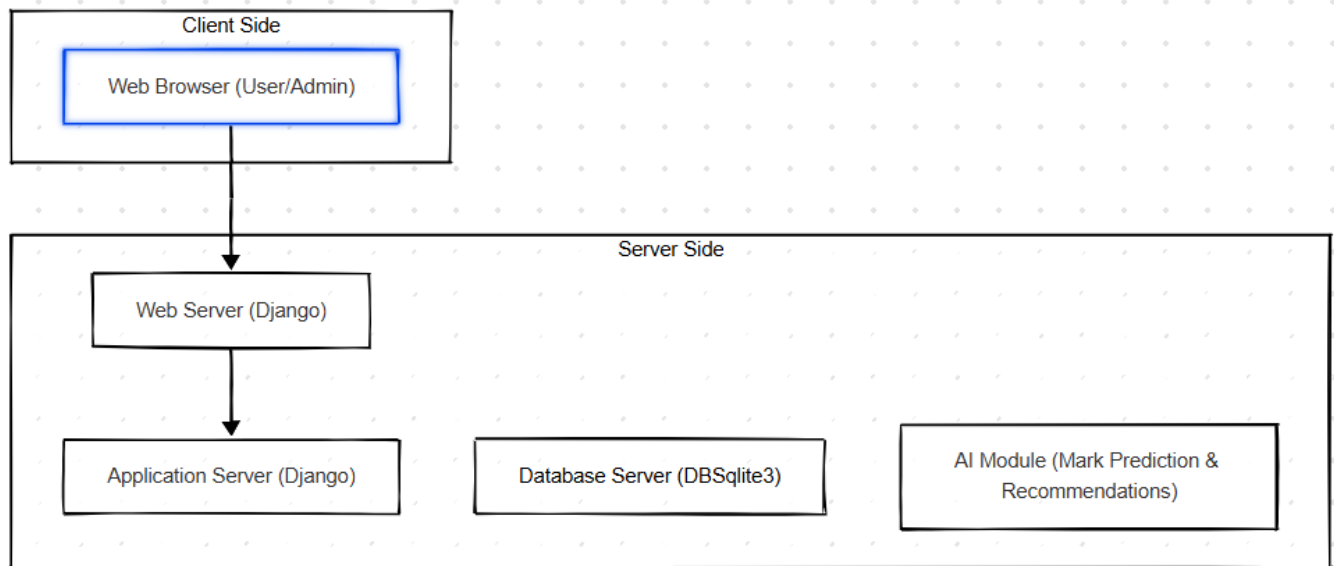
Use Case Diagram



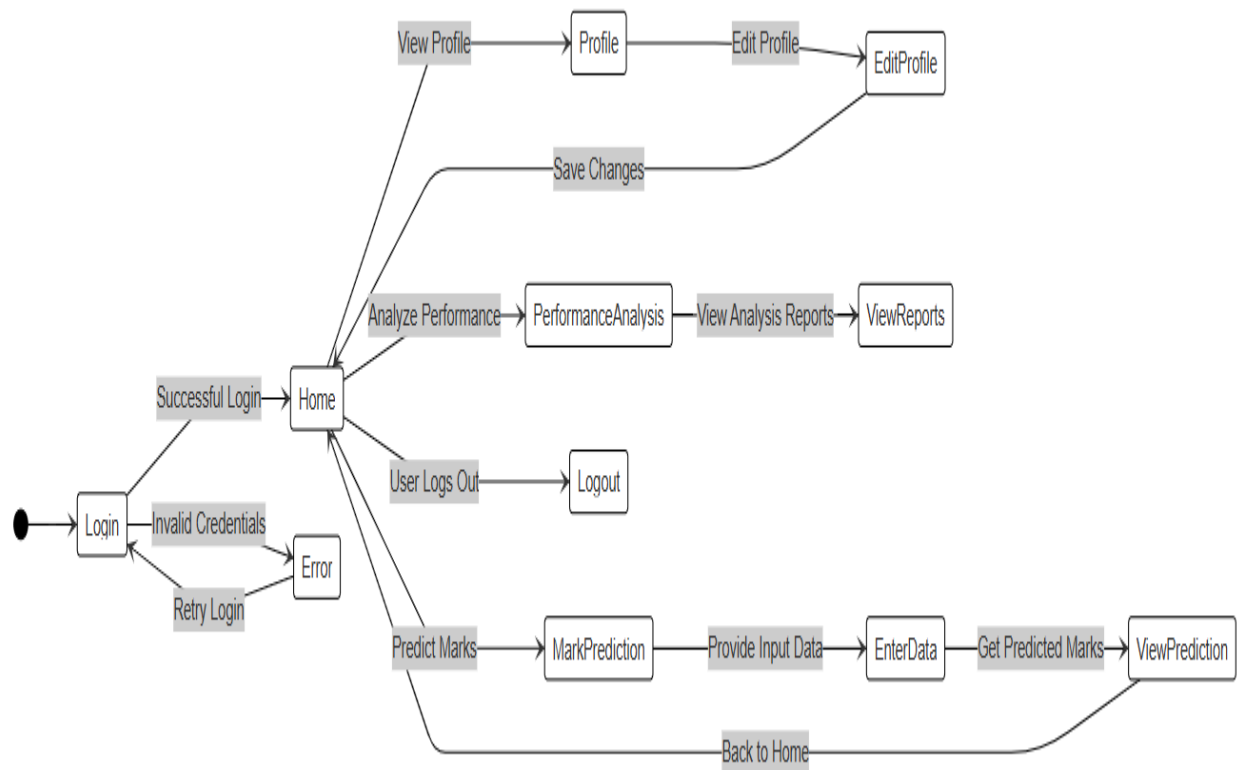
Component Diagram



Deployment Diagram



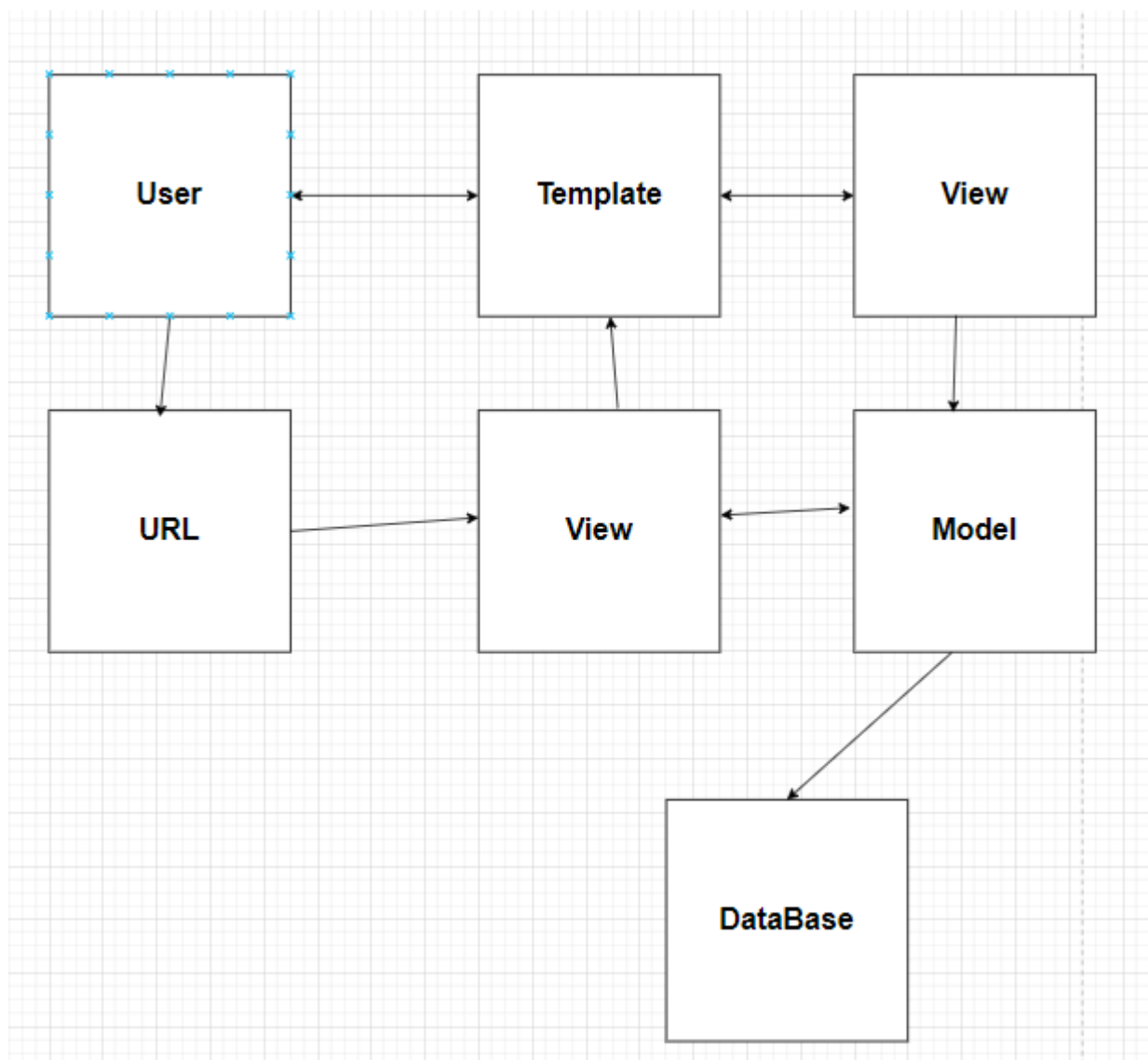
StateDiagram



Class Diagram

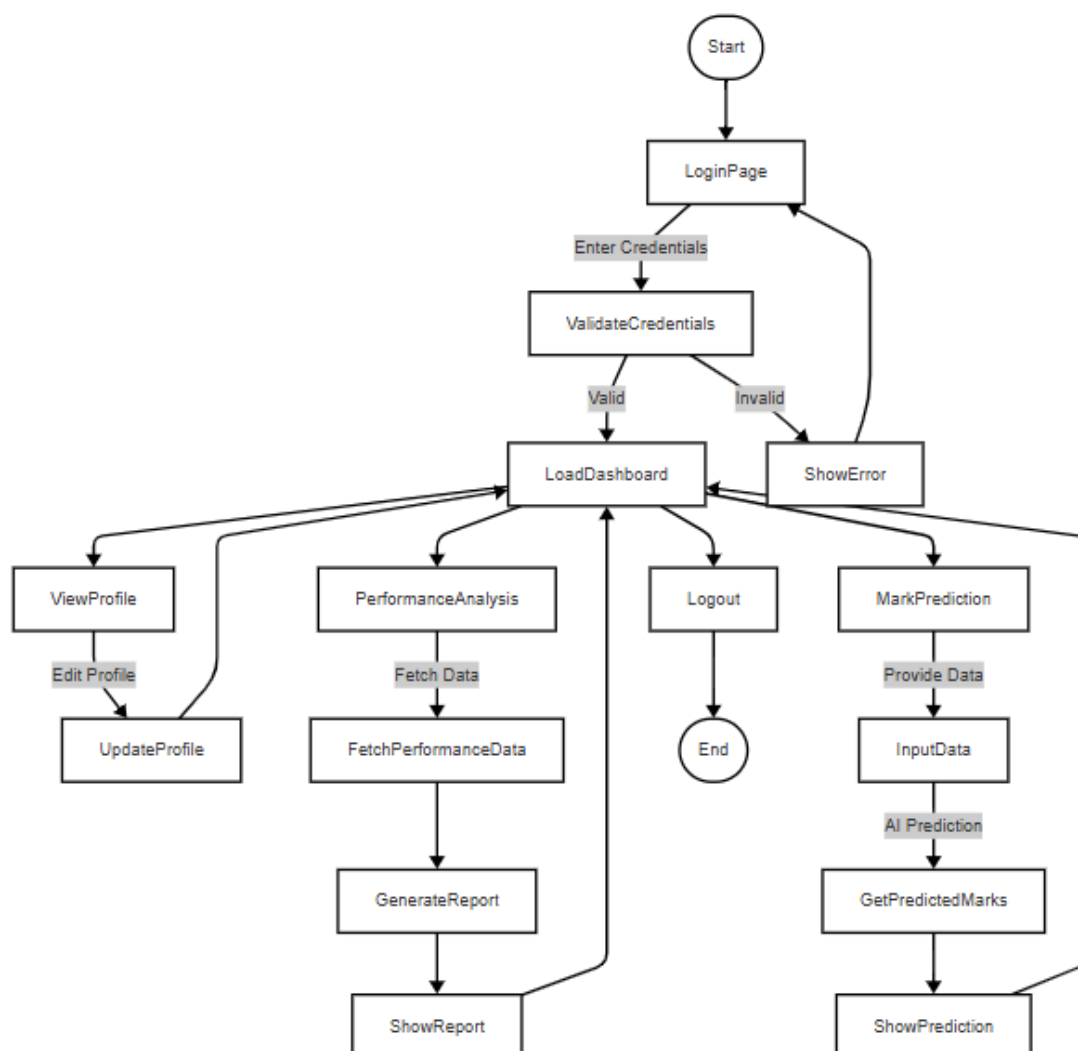


Collaboration Diagram

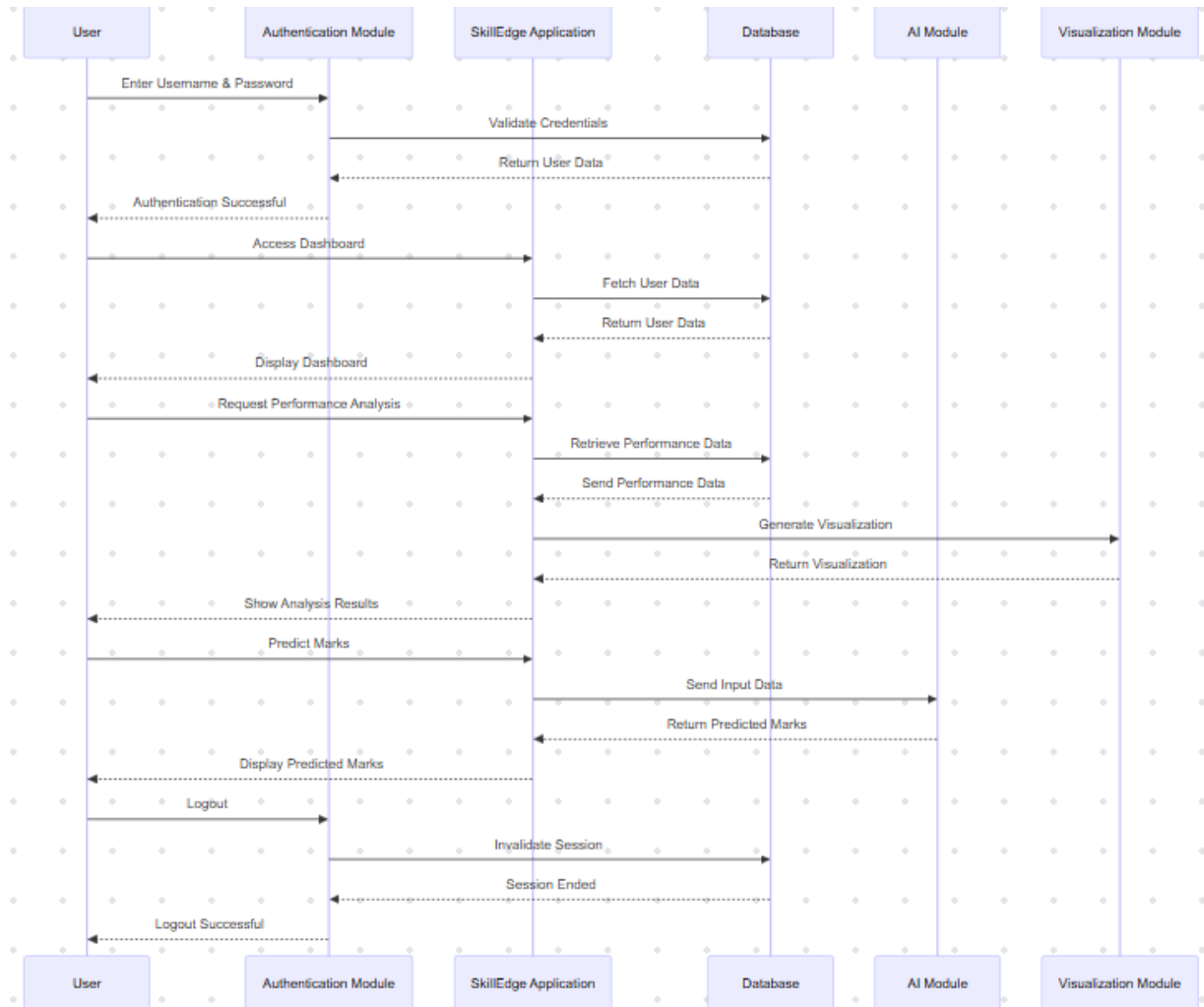


Activity Diagram

REGISTERED USER

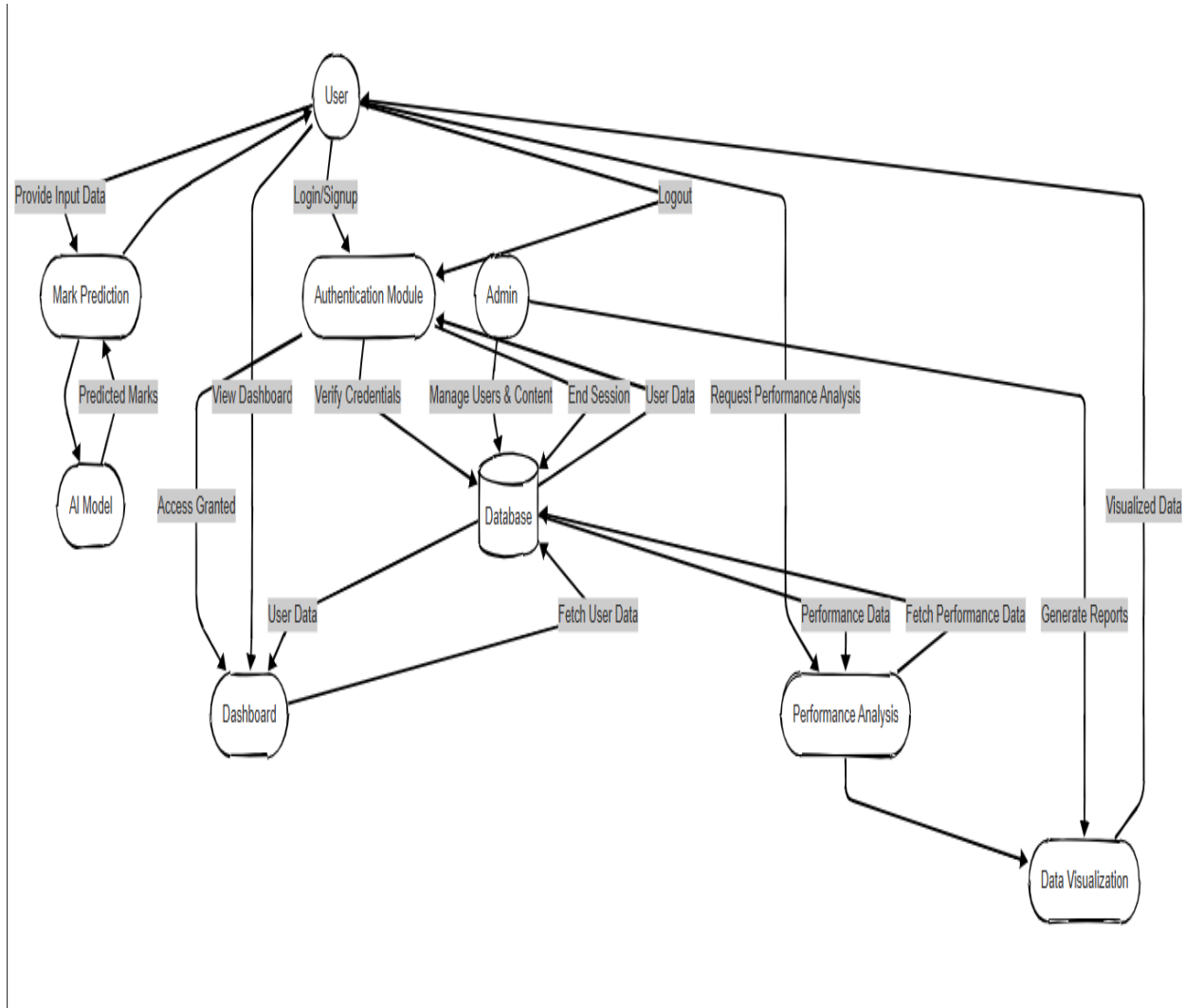


Sequence Diagram



Data Flow Diagram

Data Flow Diagram(DFD)



System Implementation

Code Implementation

Views.py

```
from django.shortcuts import render, redirect , get_object_or_404
from .models import Semester , Subject
from .forms import SemesterForm, SubjectForm
from .prediction_model import predict_next_semester_avg

def manage_performance(request):
    semesters = Semester.objects.filter(user=request.user)
    semester_form = SemesterForm()
    subject_form = SubjectForm()

    if request.method == "POST":
        if 'add_semester' in request.POST:
            semester_form = SemesterForm(request.POST)
            if semester_form.is_valid():
                semester = semester_form.save(commit=False)
                semester.user = request.user
                semester.save()
                return redirect('manage_performance')

        elif 'add_subject' in request.POST:
            subject_form = SubjectForm(request.POST)
            if subject_form.is_valid():
                subject = subject_form.save(commit=False)
                subject.semester_id = request.POST.get('semester_id') # Assign semester ID dynamically
                subject.save()
                return redirect('manage_performance')

    return render(request, 'performance/add_performance_data.html', {
        'semesters': semesters,
        'semester_form': semester_form,
        'subject_form': subject_form,
    })

def performance_dashboard(request):
    semesters = Semester.objects.filter(user=request.user).prefetch_related('subjects')

    semester_data = []
    for semester in semesters:
        subjects = semester.subjects.all()
        total_marks = sum(subject.obtained_marks for subject in subjects if subject.obtained_marks is not
None)
        max_marks = sum(subject.max_marks for subject in subjects)
        avg_marks = (total_marks / max_marks) * 100 if max_marks > 0 else 0
```

```

semester_data.append({
    'semester_name': semester.name,
    'avg_marks': avg_marks,
    'subjects': [{ 'name': subj.name, 'marks': subj.obtained_marks, 'max_marks': subj.max_marks } for
subj in subjects]
})

# Predict the next semester's average
predicted_avg = predict_next_semester_avg(request.user)

context = {
    'semester_data': semester_data,
    'predicted_avg': predicted_avg,
}

return render(request, 'performance/performance.html', context)

def delete_subject(request, subject_id):
    subject = get_object_or_404(Subject, id=subject_id)

    if request.method == "POST":
        subject.delete()
        return redirect('manage_performance')

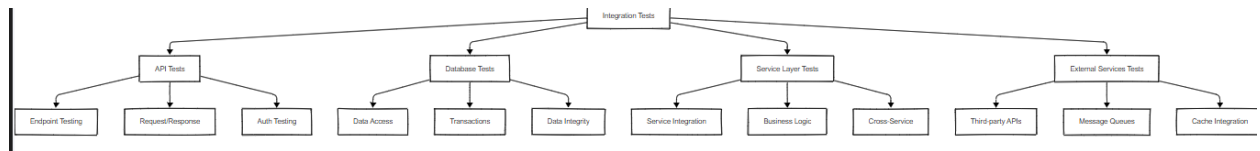
    return render(request, 'add_performance_data.html', {'subject': subject})

```

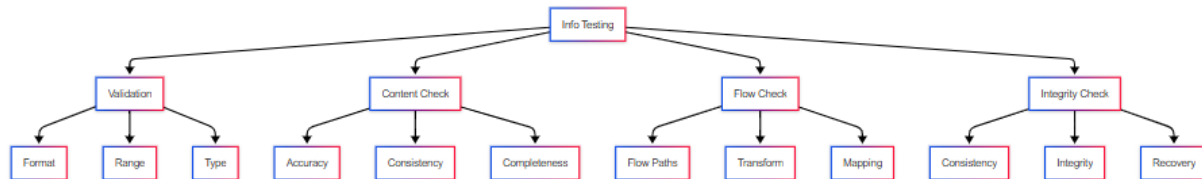
Results

Test Case

Integration Test



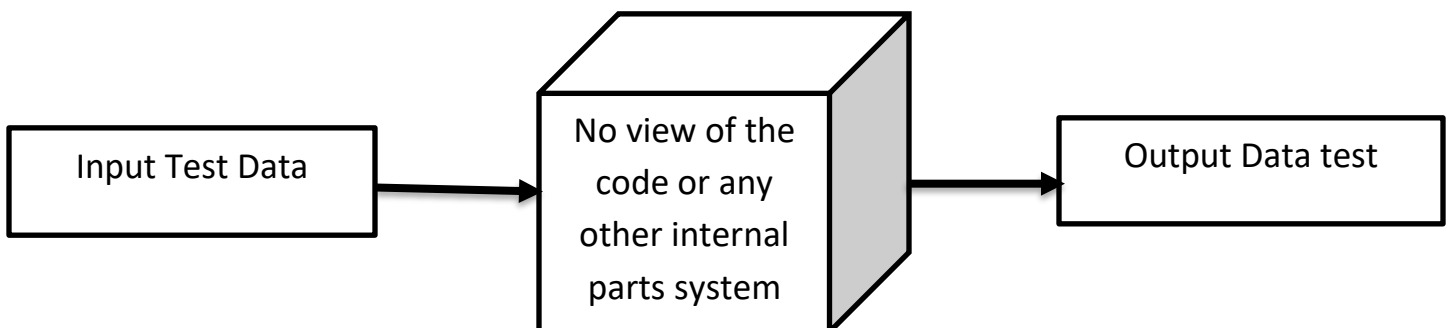
InformationTest



BlackBox Test

Black Box Testing involves only the observation of the output for certain input values, and there is no attempt to analyze the programming code that enables to arrive at those outputs. Therefore, it is also termed “data-driven, input/output driven or requirements based testing”.

Black box testing is also called as “Behavioral testing” because it only checks the system for certain inputs and doesn’t think about the internal coding. It is concerned with testing the functions of the system. Therefore, it is also called as “Functional Testing”.



DataBase Structure

USER

Field	Type	Constraints
user_id	int	Primary Key
username	string	Unique Key
email	string	Unique Key
password_hash	string	Not Null
date_joined	datetime	Not Null
is_active	boolean	Default: True
role	string	Not Null

USER_PROFILE

Field	Type	Constraints
profile_id	int	Primary Key
user_id	int	Foreign Key (USER)
first_name	string	Not Null
last_name	string	Not Null
bio	text	Nullable
profile_picture	string	Nullable
location	string	Nullable
last_active	datetime	Not Null

SKILL

Field	Type	Constraints
skill_id	int	Primary Key
name	string	Not Null
description	string	Nullable
category	string	Nullable
proficiency_level	int	Default: 0



PERFORMANCE_METRIC

Field	Type	Constraints
metric_id	int	Primary Key
user_id	int	Foreign Key (USER)
metric_type	string	Not Null
score	float	Default: 0.0
recorded_date	datetime	Not Null
feedback	text	Nullable



COMMUNITY_POST

Field	Type	Constraints
post_id	int	Primary Key
user_id	int	Foreign Key (USER)
title	string	Not Null
content	text	Not Null
created_at	datetime	Not Null
likes_count	int	Default: 0
category	string	Nullable

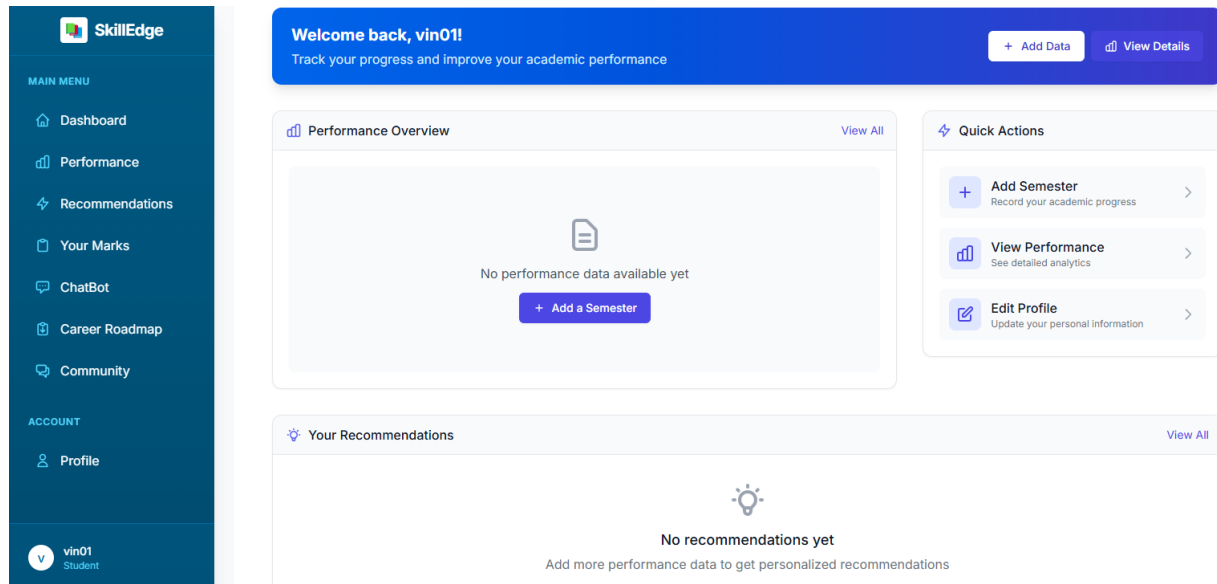


CAREER_ROADMAP

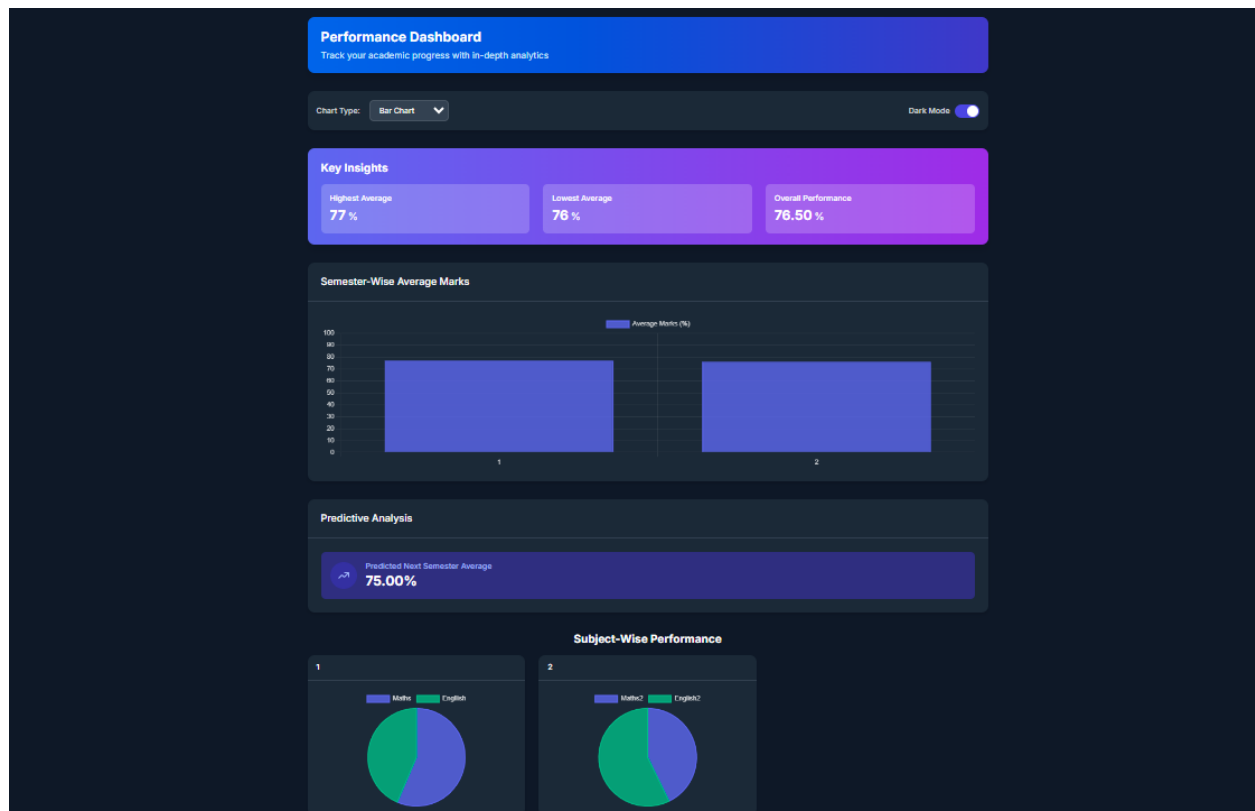
Field	Type	Constraints
roadmap_id	int	Primary Key
title	string	Not Null
industry	string	Nullable
description	text	Not Null
skill_requirements	json	Not Null
experience_level	int	Not Null

Screenshot

Main Page



Performance Page



Recommendation

Your Personalized AI Recommendations

Custom learning insights based on your performance

Refresh

English

Performance 67 / 100

Suggestion

Okay, a score of 67/100 in English suggests you're performing at a moderate level. It's not a failing grade, but there's definitely room for improvement to reach a stronger grasp of the subject. You're not particularly weak, but not exceptionally strong either. Study Tip: Focus on actively practicing grammar and sentence structure through writing. Rewrite short passages in your own words, paying close attention to word choice and clarity. Regularly reviewing common grammatical errors can improve your accuracy. Resource Links: 1. Video (Grammar): This video from mmmEnglish clearly explains common grammatical errors and how to fix them. 2. Article/Interactive Tutorial (Sentence Structure): Grammarly's blog post helps you understand sentence types and learn how to construct better sentences.

Key Resources (2)

- [YouTube Video](#)
- [Learning Resource](#)

Maths2

Performance 65 / 100

Suggestion

A score of 65/100 in Maths2 suggests you're in a moderate position. It's not a failing grade, but it also indicates there's room for improvement. You are not necessarily weak, but you're also not performing at a strong level. It means you likely understand the core concepts but struggle with more complex applications or problem-solving. Study Tip: Identify specific areas where you lost marks on the exam. Focus on re-learning those topics with a variety of examples. Then, immediately practice similar problems. Resource Links: Video (Complex Problem Solving):** (MIT OpenCourseWare's single variable Calculus section (Lecture 45) on problem solving. This covers strategic approaches to tackling tough math problems.) Interactive Tutorial (Application-Based Learning):** (Khan Academy's Differential Equations section which is used in many Maths 2 Courses. Includes practice problems, videos, and quizzes to test your understanding of complex topics and see real-world applications.)

Key Resources (2)

Chat-Bot

SkillEdge Assistant

system. While true "general AI" remains a distant goal, "narrow AI" is already having a significant impact on the world around us.

You:
how can i increase my marks

Assistant:
Okay, let's break down how to increase your marks. It's not just about studying harder, it's about studying *smarter* and addressing all the factors that contribute to your success. Here's a comprehensive plan:

I. Understanding the Problem:

Type your message...

Press Enter to send, Shift+Enter for new line

Roadmap

Career Roadmaps

Personalized career development plans based on your interests and goals

AI-generated roadmaps to help you achieve your career goals.

[+ Generate New Roadmap](#)



No roadmaps yet

Generate your first career roadmap based on your interests and goals.

[Generate Roadmap](#)

Community Post

Community

- All Posts
- + Create Post
- My Posts
- Notifications

TOPICS

Who is Elon Musk ??

[← Back to Posts](#)

V vin01 Posted 0 minutes ago General Discussion 2 views

Elon Musk

1

0 Answers

**No answers yet**
Be the first to answer this question!

Your Answer

Future Scope

Future Scope

The SkillEdge platform has significant potential for expansion and enhancement across multiple dimensions. In the immediate future, you could implement advanced AI-driven personalized learning paths that adapt to each user's learning style, pace, and career goals. The existing chatbot functionality could be enhanced with GPT-4 or similar advanced language models to provide more sophisticated learning assistance and real-time doubt resolution. A major opportunity lies in incorporating augmented reality (AR) and virtual reality (VR) components for immersive skill training, particularly beneficial for practical skills and hands-on learning experiences. From a community perspective, the platform could evolve into a comprehensive professional networking ecosystem, introducing features like peer-to-peer mentoring, expert consultation sessions, and industry-specific discussion forums.

The career roadmap feature could be enhanced with real-time labor market insights, integrating data from job markets to provide dynamic career guidance and skill requirement updates. Additionally, implementing blockchain technology for verifiable skill certificates and micro-credentials would add significant value, making the platform more attractive to both learners and employers. The recommendation system could be expanded to include predictive analytics that forecast emerging skills in various industries, helping users stay ahead of market demands. Integration with major job portals and professional networks like LinkedIn could create a seamless bridge between skill development and career opportunities. The performance tracking system could be enhanced with gamification elements, including skill badges, achievement levels, and competitive learning challenges to increase user engagement. On the technical side, implementing machine learning algorithms for early detection of learning gaps and potential dropouts could significantly improve completion rates. The platform could also benefit from mobile-first features like offline learning capabilities, microlearning modules, and push notifications for personalized learning reminders. Adding support for multiple languages and localized content would help in global expansion, while integrating with enterprise learning management systems (LMS) could open up B2B opportunities with corporations for employee upskilling programs. Lastly, developing a marketplace feature where industry experts can create and monetize their courses would add a sustainable revenue stream while ensuring fresh, relevant content.

The platform could also explore partnerships with universities and professional certification bodies to offer accredited courses and recognized certifications. These enhancements would position SkillEdge as a comprehensive, future-ready learning ecosystem that addresses the evolving needs of both individual learners and organizations in the rapidly changing digital economy.

CONCLUSION

Conclusion

SkillEdge represents a comprehensive and innovative approach to modern education and skill development in the digital age. The platform successfully integrates multiple crucial components: personalized learning paths, community engagement, career guidance, and AI-powered assistance, creating a holistic learning ecosystem. By combining traditional educational methodologies with cutting-edge technology, SkillEdge addresses the growing demand for flexible, accessible, and relevant skill development opportunities. The platform's strength lies in its multi-faceted approach, offering users not just courses but a complete career development journey.

The integration of AI-powered chatbot assistance, performance tracking, and community features creates an engaging and supportive learning environment. The career roadmap functionality provides clear direction and purpose to users' learning journeys, while the recommendation system ensures personalized and relevant content delivery. The community aspect of SkillEdge fosters peer learning and networking opportunities, essential components in today's interconnected professional world. The performance tracking system enables users to monitor their progress and maintain motivation throughout their learning journey. Furthermore, the platform's scalable architecture ensures it can evolve with technological advancements and changing educational needs. As the job market continues to evolve rapidly, SkillEdge's focus on practical skills and industry relevance positions it as a valuable tool for continuous learning and professional development.

The platform's potential for future enhancements, including AR/VR integration, blockchain certifications, and expanded AI capabilities, ensures its long-term viability and relevance in the educational technology landscape. In conclusion, SkillEdge stands as a testament to the power of combining education, technology, and community to create a meaningful impact on users' professional growth and career development. Its comprehensive approach to skill development, coupled with its future-ready architecture, makes it a significant contribution to the future of education and professional training.

REFERENCES

References:

I referred to the following books and google for completion of my project:

INTERNET REFERENCES:

Reference taken from Websites like

Django :

Official Django documentation was a primary resource for understanding various Django features and best practices for web development.

URL: <https://docs.djangoproject.com>

Stack Overflow:

Stack Overflow served as a valuable resource for troubleshooting and problem-solving during development.

URL: <https://stackoverflow.com>

ChatGPT :

I used ChatGPT for Advance Problems and Bugs

URL: <https://chatgpt.com>

Cursor AI

I used Cursor AI for my frontend and Deployment

URL: <https://www.cursor.com/>